

BANARAS HINDU UNIVERSITY

B.A./B.Sc. (Hons.) Semester V Examination, 2022-23

Paper No. MTB-606: Complex Analysis

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Symbols have their usual meanings.*

Question 1.

- (a) Let $z = (x, y)$ be a non-zero complex number. Show that the multiplicative inverse of z is:

$$z^{-1} = \left(\frac{x}{x^2 + y^2}, \frac{-y}{x^2 + y^2} \right) \quad [2]$$

- (b) Let $z_1, z_2 \in \mathbb{C}$. Show that $|z_1 - z_2| \geq ||z_1| - |z_2||$. [2]

- (c) Define the stereographic projection of $\mathbb{C}$ on $S^2 \setminus \{(0, 0, 1)\}$. Find its equations. [2]

- (d) Define $f(z) = z/\bar{z}$ for all $z \in \mathbb{C} \setminus \{0\}$. Discuss the existence of the limit $\lim_{z \rightarrow 0} f(z)$. [2]

- (e) Find the radius of convergence of the power series:

$$\sum_{n=0}^{\infty} \frac{z^n}{n!} \quad [2]$$

- (f) Let S be a non-empty subset of $\mathbb{C}$ with $\text{int}(S) \neq \emptyset$, and $f : S \rightarrow \mathbb{C}$ be a function. Explain the differentiability of f at $z_0 \in \text{int}(S)$. [2]

- (g) Define a harmonic function and give two examples of harmonic functions. [2]

- (h) Discuss polygonally connected sets and open connected sets. Write two sets which are polygonally connected in $\mathbb{C}$. [2]

- (i) Let $f : [a, b] \rightarrow \mathbb{C}$ be a continuous complex-valued function. Show that:

$$\left| \int_a^b f(t) dt \right| \leq \int_a^b |f(t)| dt \quad [2]$$

- (j) Evaluate the integral $\int_C dz$, where $C = \{\gamma(t) : t \in [a, b]\}$ is any rectifiable curve with $\gamma(a) = \alpha$ and $\gamma(b) = \beta$. [2]

- (k) Evaluate $\int_C \frac{e^z}{z(z-2)} dz$, where C is the boundary of the annulus between the circles $|z| = 1$ and $|z| = 3$. [2]

Question 2.

- (a) Let $f(z) = |z|^2$ for all $z \in \mathbb{C}$. Prove that f is differentiable only at $z = 0$. Also, show that f is nowhere analytic. [4]
- (b) Let $f(z) = u(x, y) + iv(x, y)$ be analytic in $D = \{z \in \mathbb{C} : |z| < 1\}$ such that its real part is $u(x, y) = x^3 - 3xy^2$. Find the imaginary part $v(x, y)$ of $f(z)$. Furthermore, find $f(z)$ in terms of z . [5]
- (c) Discuss n -th roots of a complex number z . Hence find the fifth roots of $(1 + i)$. [5]

Question 3.

- (a) Let D be a non-empty subset of $\mathbb{C}$ and $z_0 \in D$. Let $f, g : D \rightarrow \mathbb{C}$ be functions such that $\lim_{z \rightarrow z_0} f(z) = w_0$ and $\lim_{z \rightarrow z_0} g(z) = W_0$. Show that:

$$\lim_{z \rightarrow z_0} f(z)g(z) = w_0W_0$$

[4]

- (b) Suppose that $f(z) = u(x, y) + iv(x, y)$, $z = x + iy$, and $z_0 = x_0 + iy_0$, $w_0 = u_0 + iv_0$. Show that:

$$\lim_{z \rightarrow z_0} f(z) = w_0 \iff \lim_{(x,y) \rightarrow (x_0,y_0)} u(x, y) = u_0 \text{ and } \lim_{(x,y) \rightarrow (x_0,y_0)} v(x, y) = v_0$$

[4]

- (c) Discuss the relation between continuity, differentiability and analyticity of a complex variable function at a point of its domain. Explain the implications between them and give supportive examples. [6]

Question 4.

- (a) Let D be a non-empty open set in $\mathbb{C}$ and $z_0 = x_0 + iy_0 \in D$. Let the function $f(z) = u(x, y) + iv(x, y)$ for all $z = x + iy \in D$. Suppose that:

- (i) the first-order partial derivatives of u and v with respect to x and y exist everywhere in a neighborhood of z_0 ;
- (ii) these partial derivatives are continuous at (x_0, y_0) and satisfy the Cauchy-Riemann equations: $u_x = v_y$ and $u_y = -v_x$ at (x_0, y_0) .

Show that $f'(z_0)$ exists and $f'(z_0) = u_x(x_0, y_0) + iv_x(x_0, y_0)$. [5]

- (b) State and prove Cauchy Integral Formula. [5]

- (c) Let $f(z) = u + iv$ be an analytic function. Show that

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) |f(z)|^2 = 4|f'(z)|^2$$

[4]

Question 5.

- (a) Suppose that a function $f(z) = u(x, y) + iv(x, y)$ and its conjugate $\overline{f(z)} = u(x, y) - iv(x, y)$ are defined in a given domain D such that both $f(z)$ and $\overline{f(z)}$ are analytic in D . Show that $f(z)$ is constant in D . [5]
- (b) Let f be an analytic function in a simply connected domain D . Then, along any simple closed contour C in D , show that

$$\int_C f(z) dz = 0$$

[5]

- (c) State and prove Liouville's theorem. [4]

Question 6.

- (a) State and prove the Fundamental Theorem of Algebra. [6]
- (b) Evaluate

$$\int_C \frac{1}{(z-a)^m} dz, \quad m = 1, 2, \dots, N$$

where C is a simple closed contour containing the point a inside it. [8]

Question 7.

- (a) State and prove Taylor's theorem for analytic function of complex variable. [7]
- (b) Find all possible Taylor's and Laurent series expansions of the function

$$f(z) = \frac{1}{(z+1)(z+3)}$$

about the point $z = 1$. [7]